Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Problem Statement

In modern organizations, large volumes of data are frequently imported from external sources such as spreadsheets, databases, and third-party applications into platforms like ServiceNow. However, manual data import processes or poorly configured data loading mechanisms often result in inconsistencies, duplication, and incorrect relationship mapping between records. This leads to challenges in maintaining data integrity, especially when dealing with interconnected tables such as users, departments, and assets. Additionally, the lack of automation increases processing time and the likelihood of human error. Therefore, there is a need for an efficient and automated solution that can accurately import data and establish proper relationships between records using mechanisms like Import Sets and Dot Walking.

Objectives :

The primary objective of this project is to design and implement an automated data import process that ensures accuracy, consistency, and efficiency within the ServiceNow platform. This includes utilizing Import Sets and Transform Maps to systematically convert raw data into structured records while minimizing duplication and errors. Another key goal is to leverage Dot Walking techniques to correctly establish and maintain relationships between related tables. The project also aims to

reduce manual intervention, improve data quality, and create a scalable and reusable framework for handling recurring data imports across different business use cases.

3. Required Skills

To successfully implement this, you should have:

Technical Skills

- Knowledge of ServiceNow platform
- Understanding of:
  - Import Sets
  - Transform Maps
  - Coalesce fields
- Familiarity with **Dot Walking** (navigating reference fields)
- Basic scripting (JavaScript in ServiceNow)
- Database concepts (tables, relationships, foreign keys)

Analytical Skills

- Data mapping and transformation logic
- Problem-solving for data inconsistencies
- Debugging transform errors

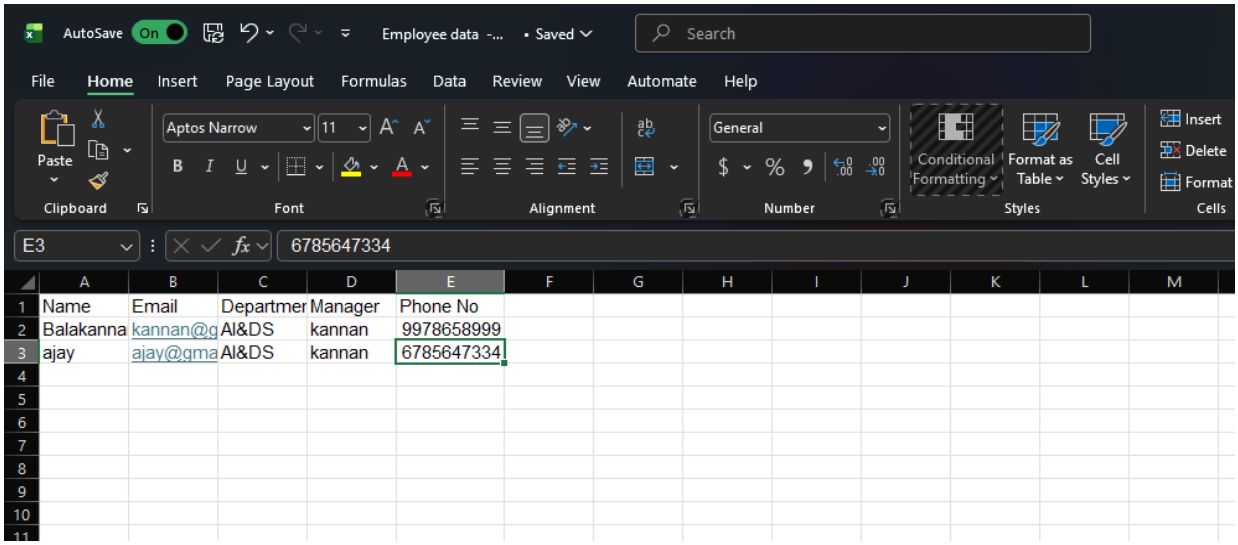
Tools & Concepts

- CSV / Excel data handling
- Data validation techniques
- Workflow automation basics

4. Task Initialization (Step-by-Step)

Step 1: Create records in the Sheet

1. Open google sheet

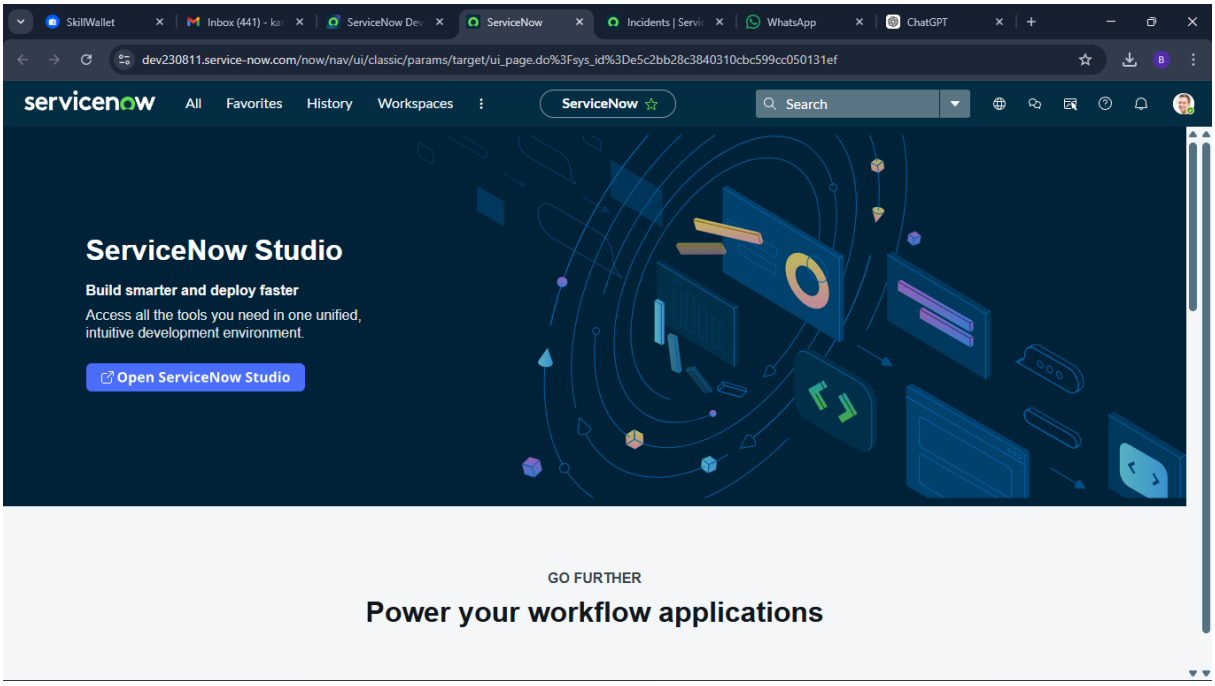


2. Name the fields as name,email,department, manager and phone number
3. Enter some records
4. Save the sheet as: employee data
5. Download the file as .xlsx , .csv (or) .xml file

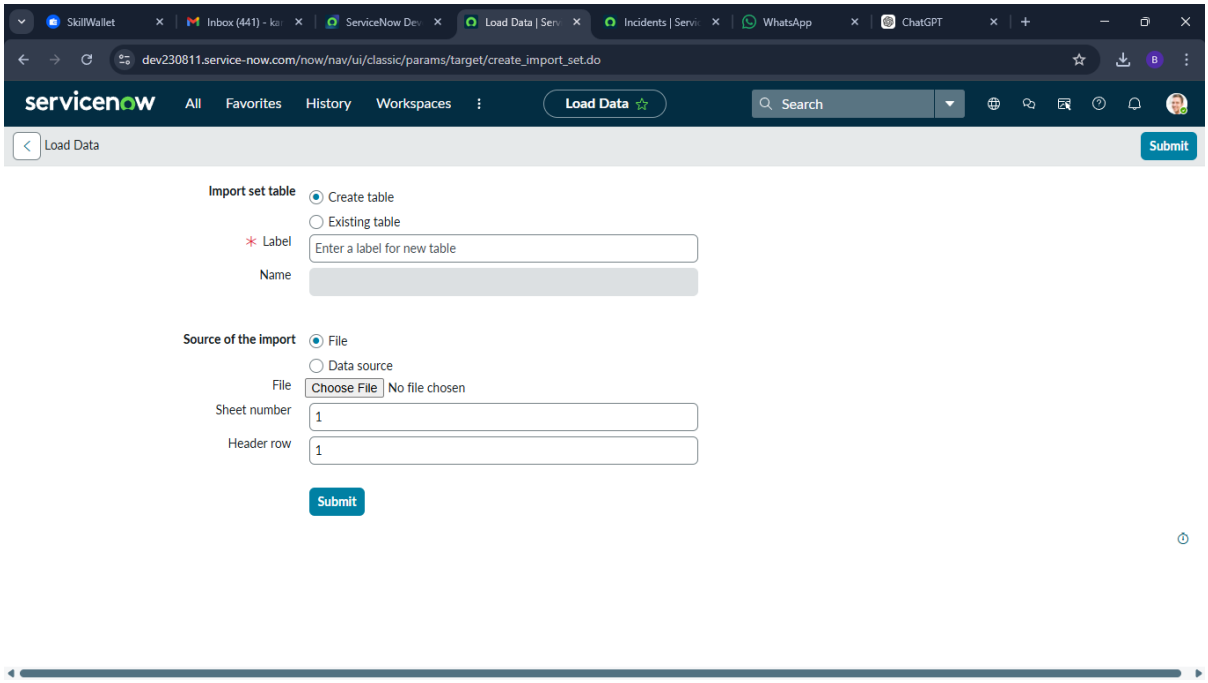
Step 2: Import data

Creating Import Table(Staging Table)

1. Open service now.

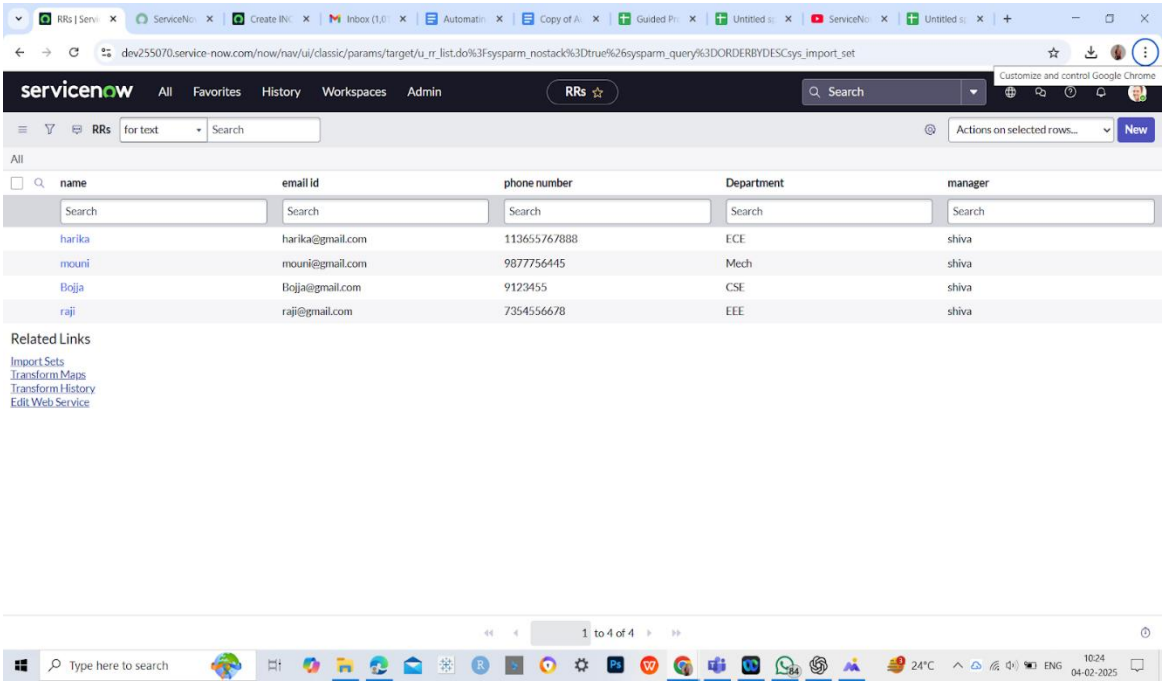


2. Click on All >> search for import sets



3. Select load data under system import sets
4. select create table
5. Give the the import set table name as: import employee
6. Choose the file which we created in the XL sheet.

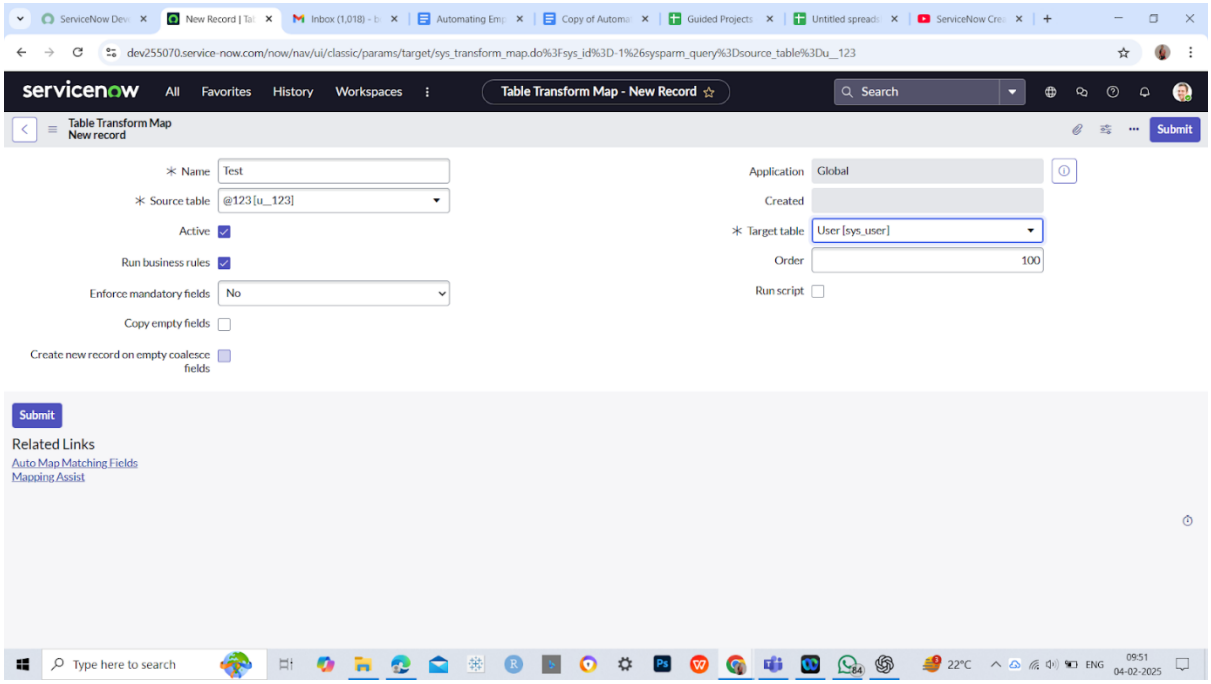
7. The data is imported,click on personalized list



8. Add fields to the form.

Step 3: Create transform map

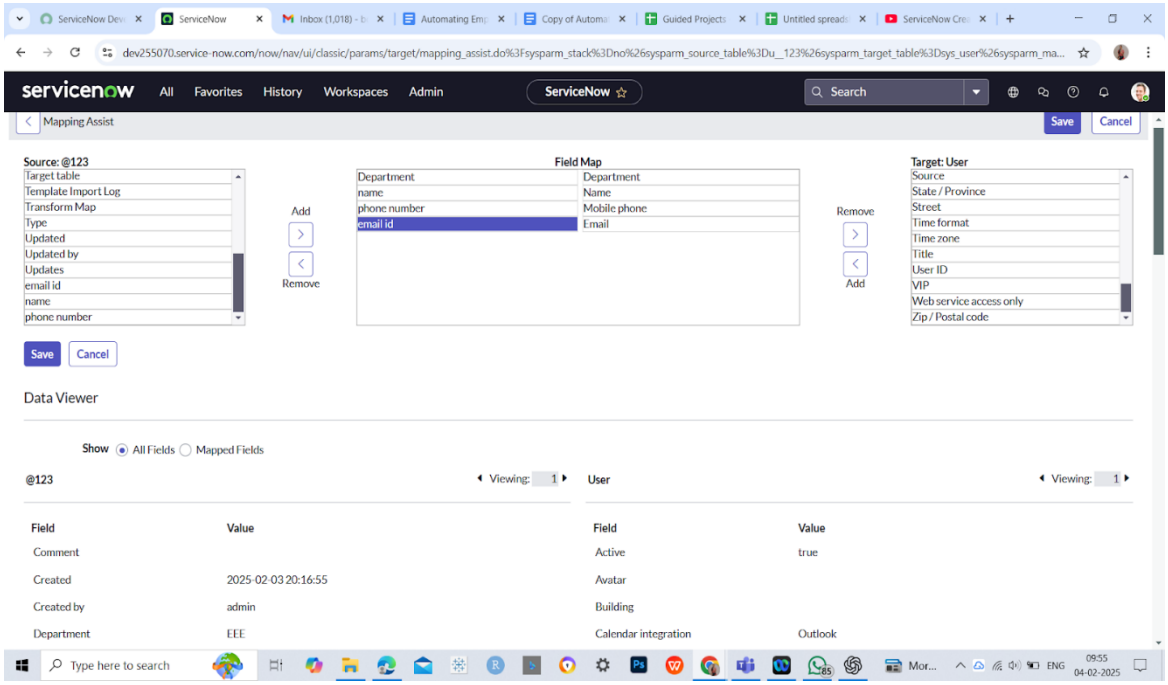
1. Open service now.



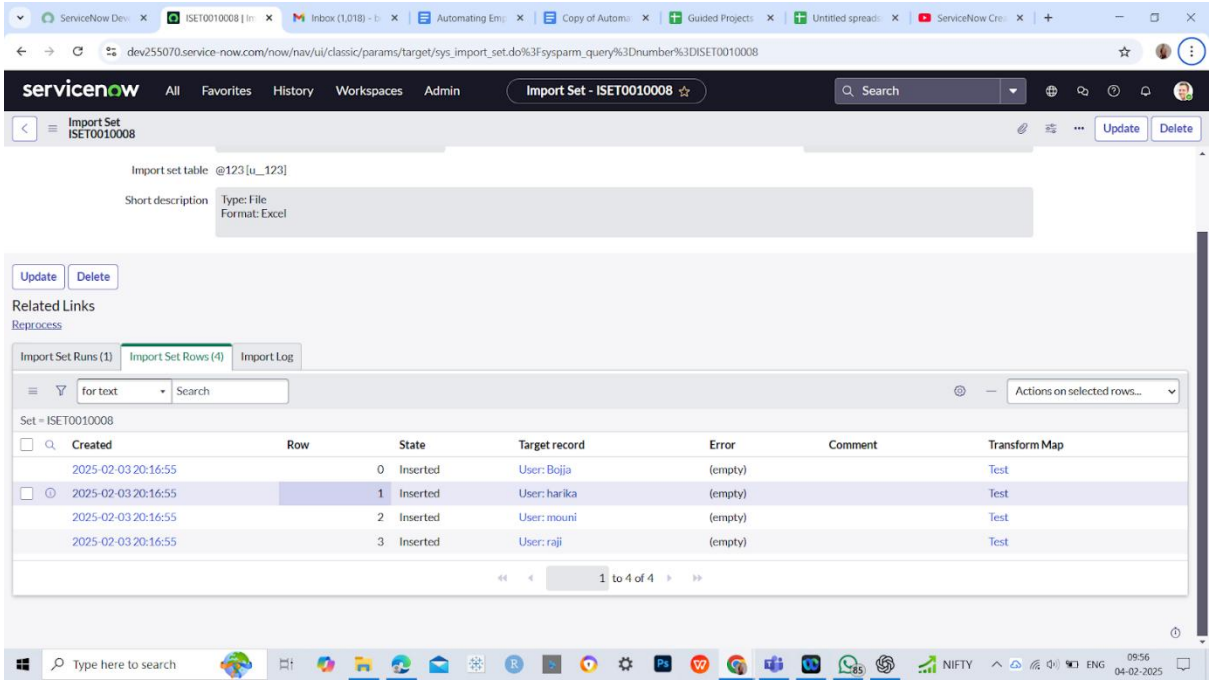
2. Click on All >> search for import set
3. Select create transform map under import sets
4. Click on new
5. Fill the following details to create a new transform map
6. Give name as: Test
7. Select source table: @123
8. Select target table: user[sys_user]
9. Save the form.

Step 4: Match the fields using mapping assist

1. Scrolled down the created transform map form
2. Click on auto matching fields



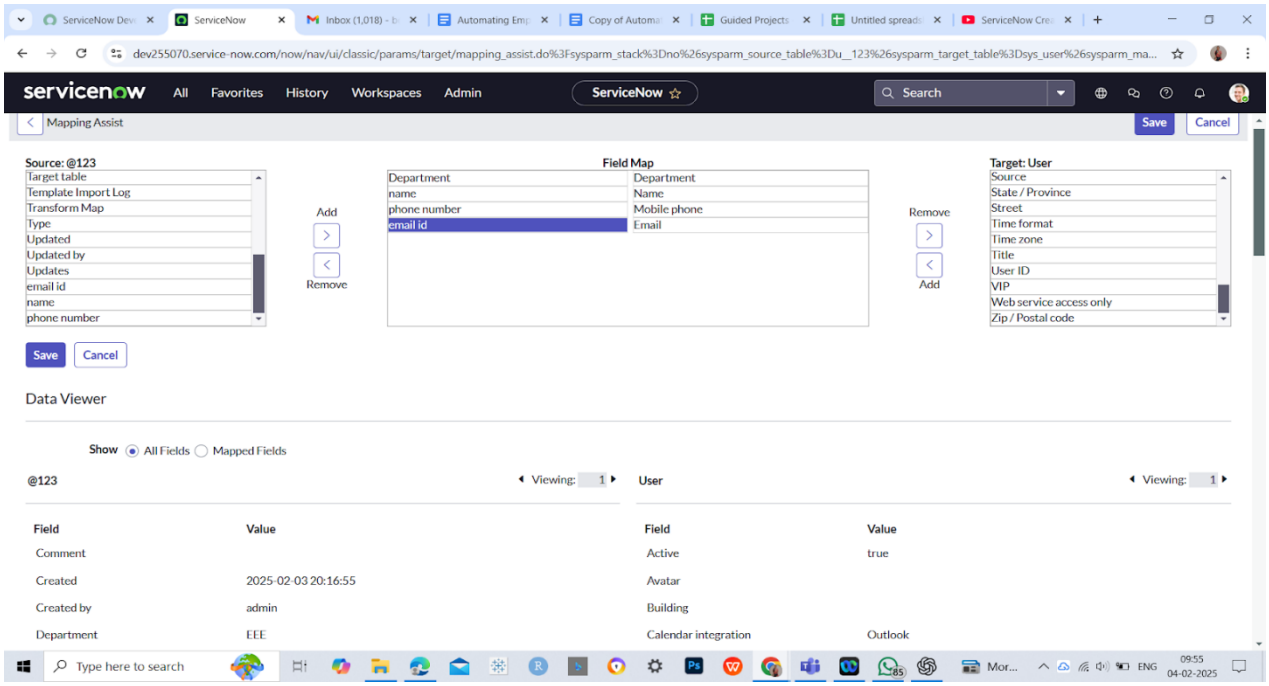
3. Then in the mapping fields some fields are matched
4. Using mapping assist map the manager field.
5. Select the fields which you are mapping in the staging table and target table
6. Select manager in source table and add



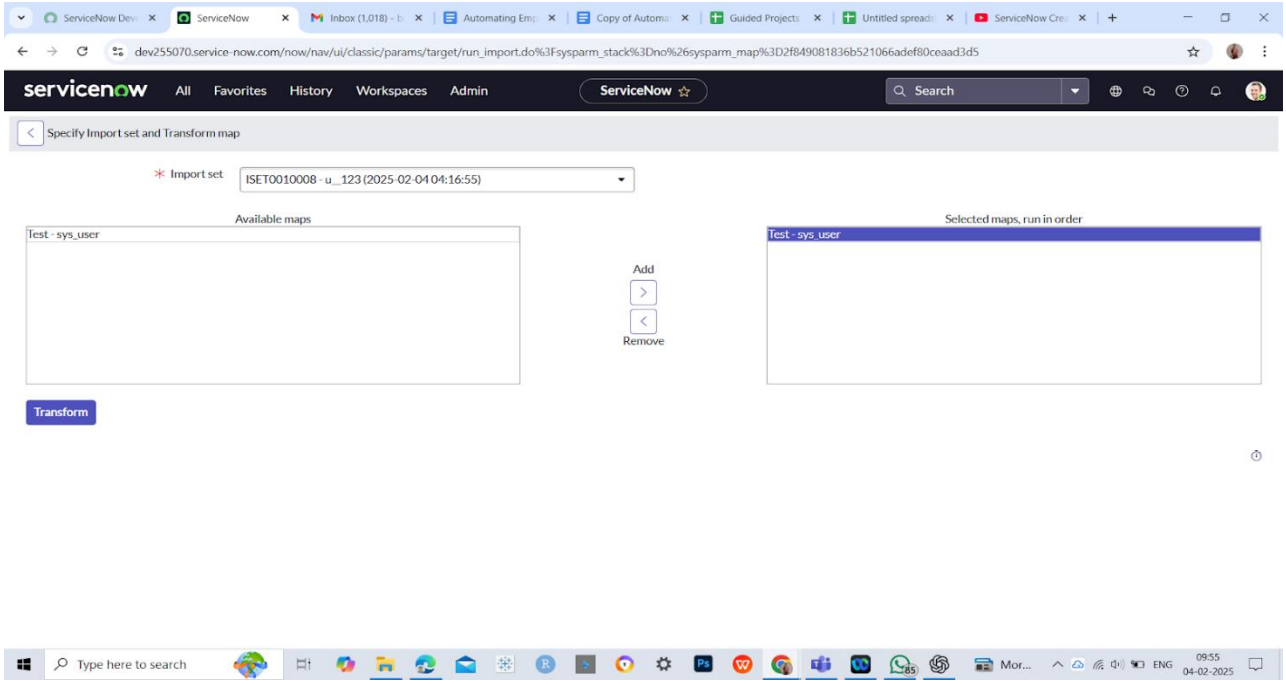
7. Select manager field in target table
8. Click on the save button.

Step 5: Transform the Data

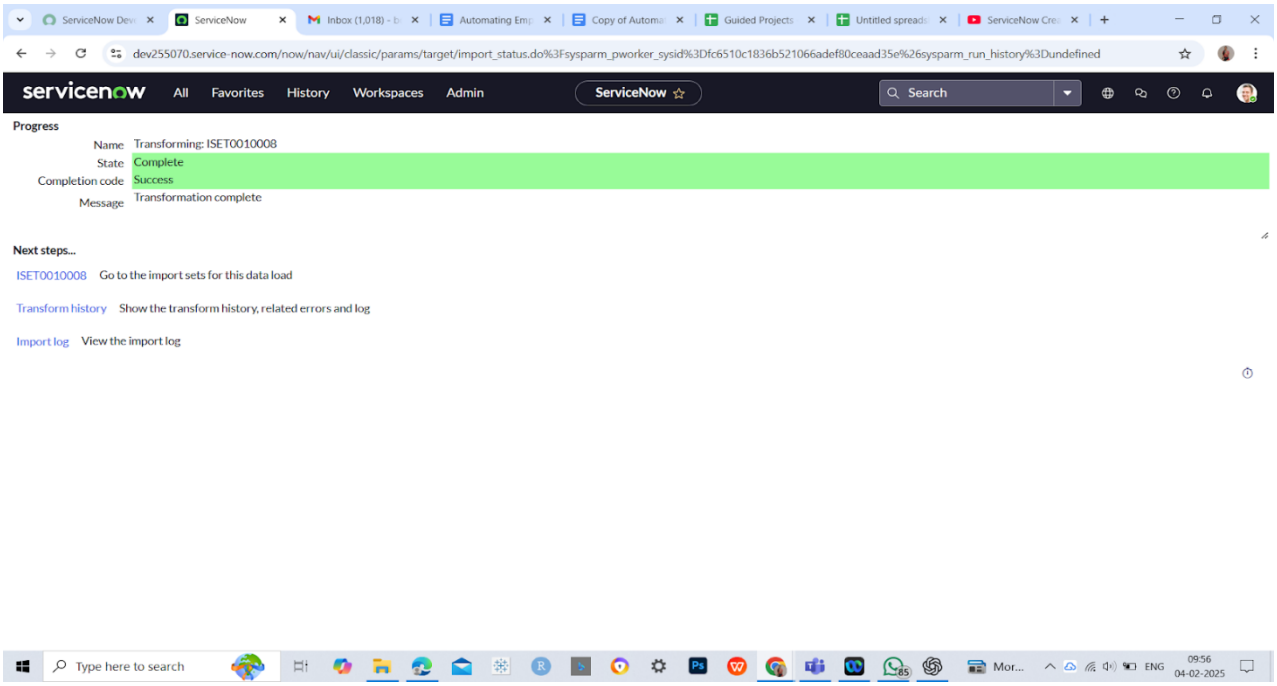
1. Click on transform in the related links.



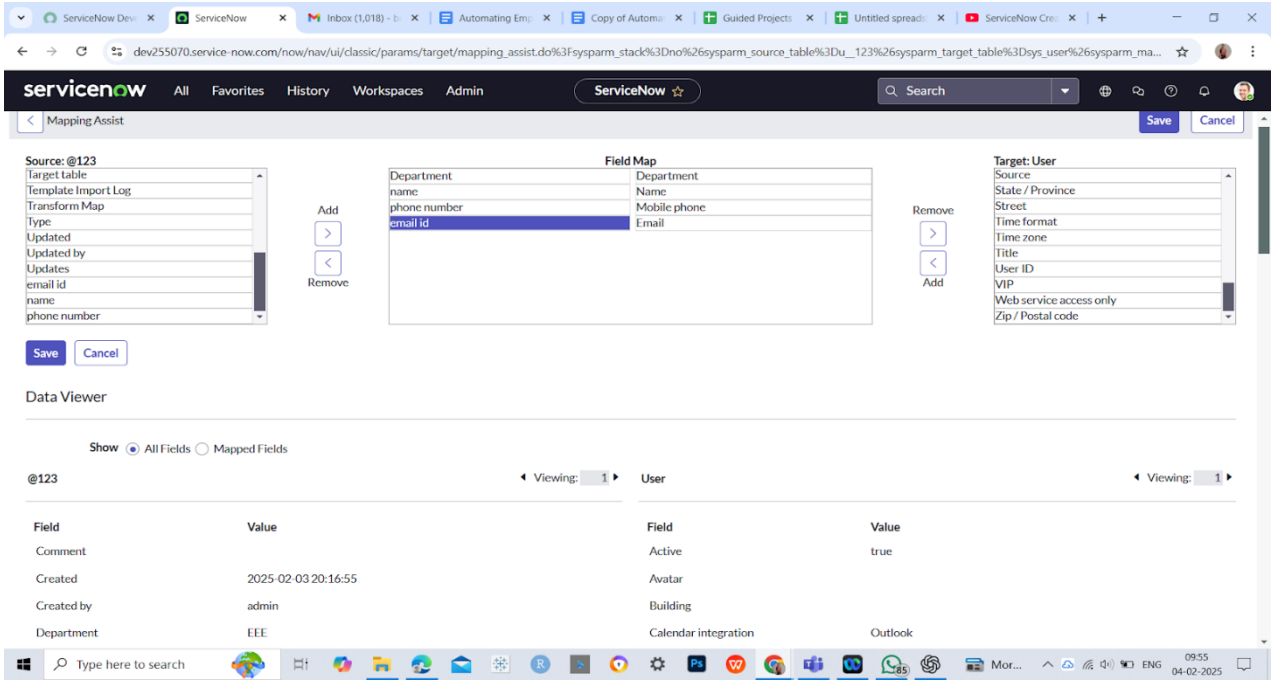
2. Click on transform.

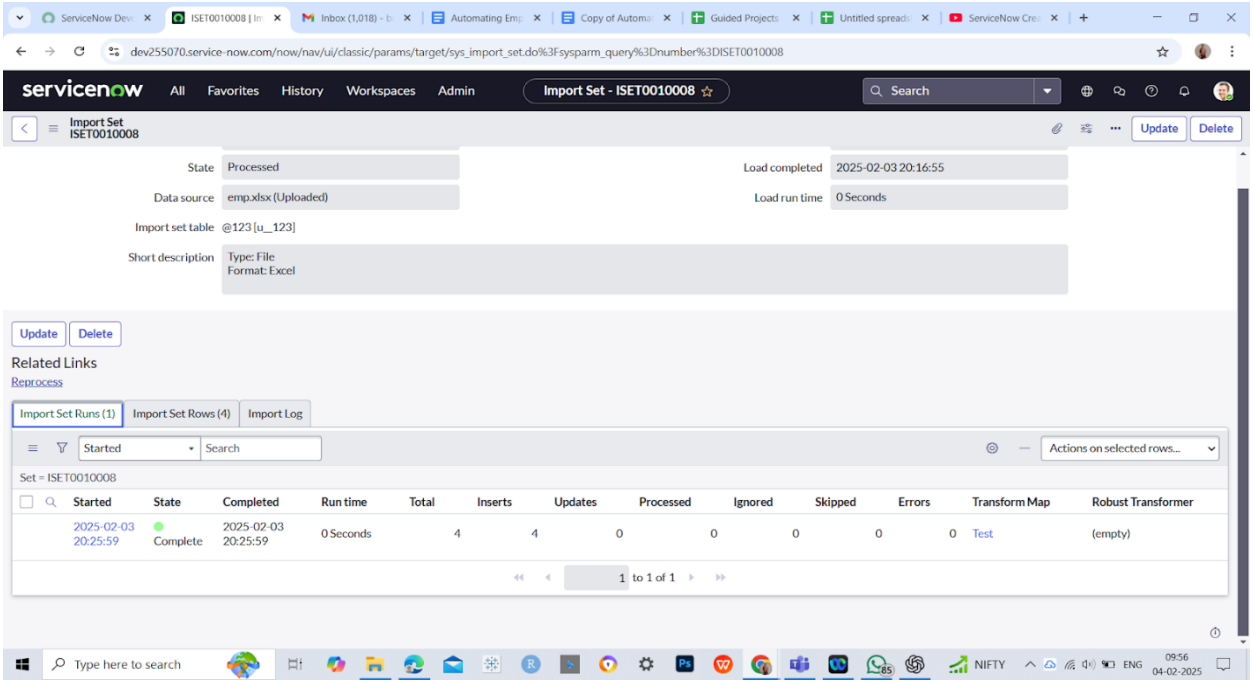


3. Data's transform state is completed.



4. Click on ISET0010008 under next steps





5. Four rows are inserted.

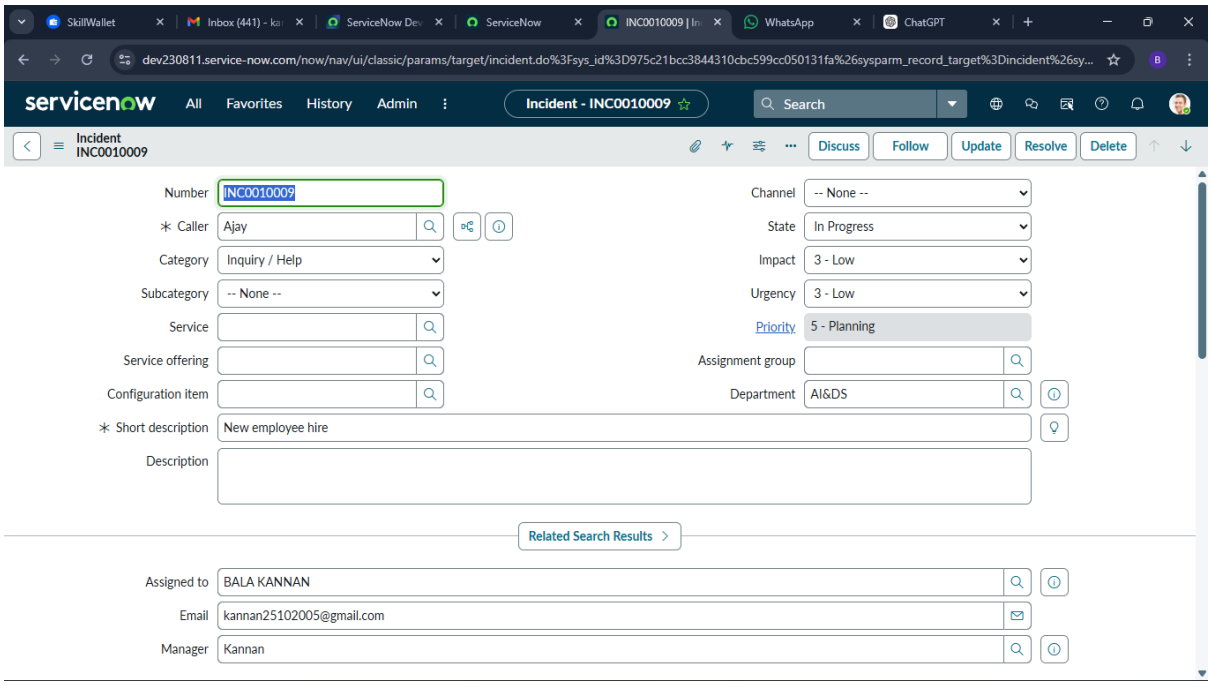
Dot walking

Dot Walking in **ServiceNow** is a technique used to access fields from a related table using a reference field.

when one table has a **reference field** pointing to another table, dot walking allows you to retrieve fields from that referenced record without writing complex queries.

Testing

Testing is the process of verifying that a system, feature, or functionality works as expected and meets defined requirements before it is released to users.



Conclusion

This project automates data imports using Import Sets, Transform Maps, and Dot Walking in ServiceNow PDI, enabling efficient data migration and relationship mapping.

Import Sets allow easy data import from external sources.

Transform Maps ensure data is correctly mapped and processed.

Dot Walking enables automatic relationship mapping between tables.